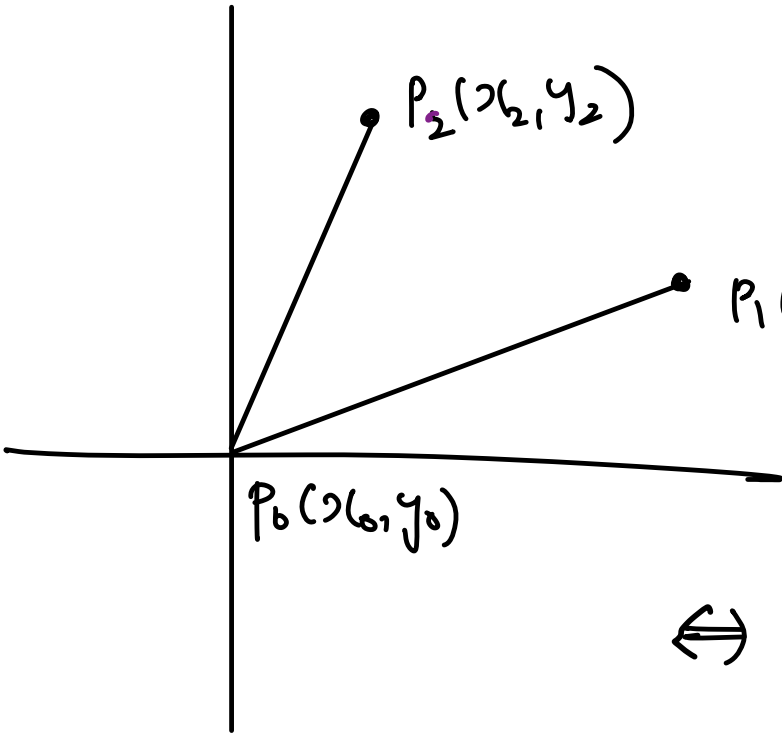
Convex Hull :



P_2 is on left of P_1
(Counter clockwise direction)

$\Leftrightarrow$ Slope (P_0P_1) < Slope (P_0P_2)

$$\text{Slope}(P_0P_1) < \text{Slope}(P_0P_2)$$

$$\frac{y_1 - y_0}{x_1 - x_0} < \frac{y_2 - y_0}{x_2 - x_0}$$

$$\Leftrightarrow (y_1 - y_0)(x_2 - x_0) < (y_2 - y_0)(x_1 - x_0)$$

$$\Leftrightarrow (y_2 - y_0)(x_1 - x_0) - (y_1 - y_0)(x_2 - x_0) > 0$$

$$\Leftrightarrow \begin{vmatrix} x_1 - x_0 & x_2 - x_0 \\ y_1 - y_0 & y_2 - y_0 \end{vmatrix} > 0$$

P_2 is on left of P_1

$$\Leftrightarrow \begin{vmatrix} x_1 - x_0 & x_2 - x_0 \\ y_1 - y_0 & y_2 - y_0 \end{vmatrix} > 0 .$$

Graham's Algorithm:

Graham-Scan (P)

1. Let p_0 be the point in P with min y-coordinate (left most in case of tie)

2. Let $\langle p_1, p_2, \ldots, p_m \rangle$ be the remaining points in P sorted by polar angle in counter clockwise order

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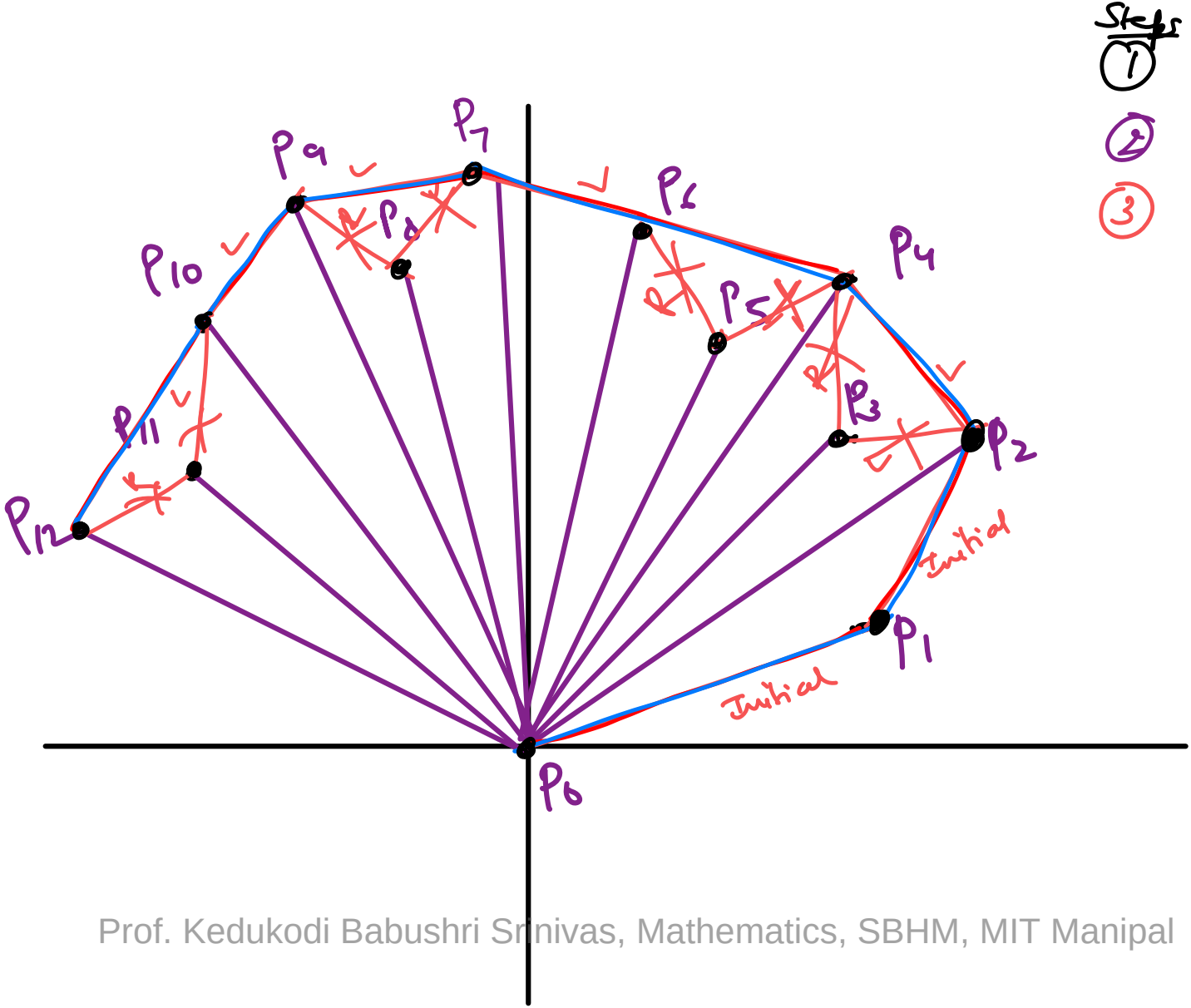
w.r.t p_0

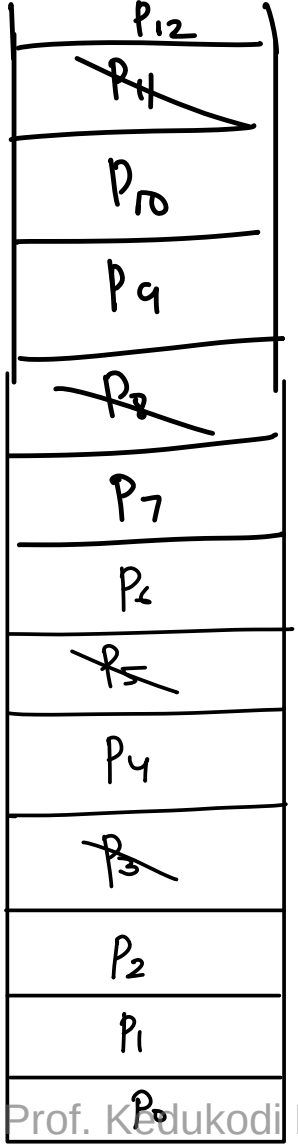
3. Push (p_0, S)
4. push (p_1, S)
5. push (p_2, S)
6. For $i \leftarrow 3$ to m
7. do while the angle formed by points Next-To-Top(S), Top(S) and p_i makes a non-left turn
8. do Pop(S)

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9. push (p_i, S)

10. return Convex Hull S

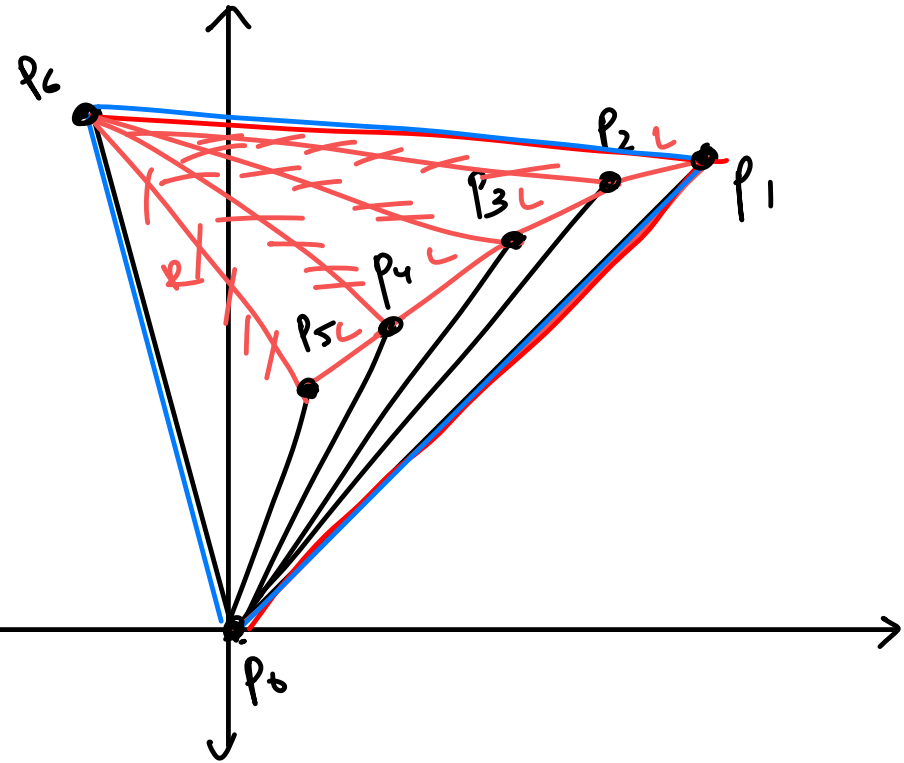

Steps
1
2
3
P_0
P_1
P_2
P_3
P_4
P_5
P_6
P_7
P_8
P_9
P_{10}
P_{11}
P_{12}
Initial
Initial

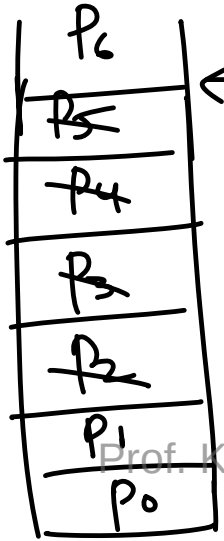


Convex Hull

$$H = \{p_0, p_1, p_2, p_4, p_6, p_7, p_9, p_{10}, p_{12}\}$$

(2)





$$M = \{P_0, P_1, P_6\}$$

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Reduction :

$A \leq_p B$ ↘

(i) B is atleast as hard as A.

(ii) B is solvable in polynomial time $\Rightarrow$ A is solvable in polynomial time.

Q: Sorting $\leq$ Convex-Hull

P → polynomial time

(Sorting is reducible from (to) Convex Hull)

Proof:

Suppose we know how to find (2D)

Convex Hull for points $p_0, p_1, \cdots, p_n$.

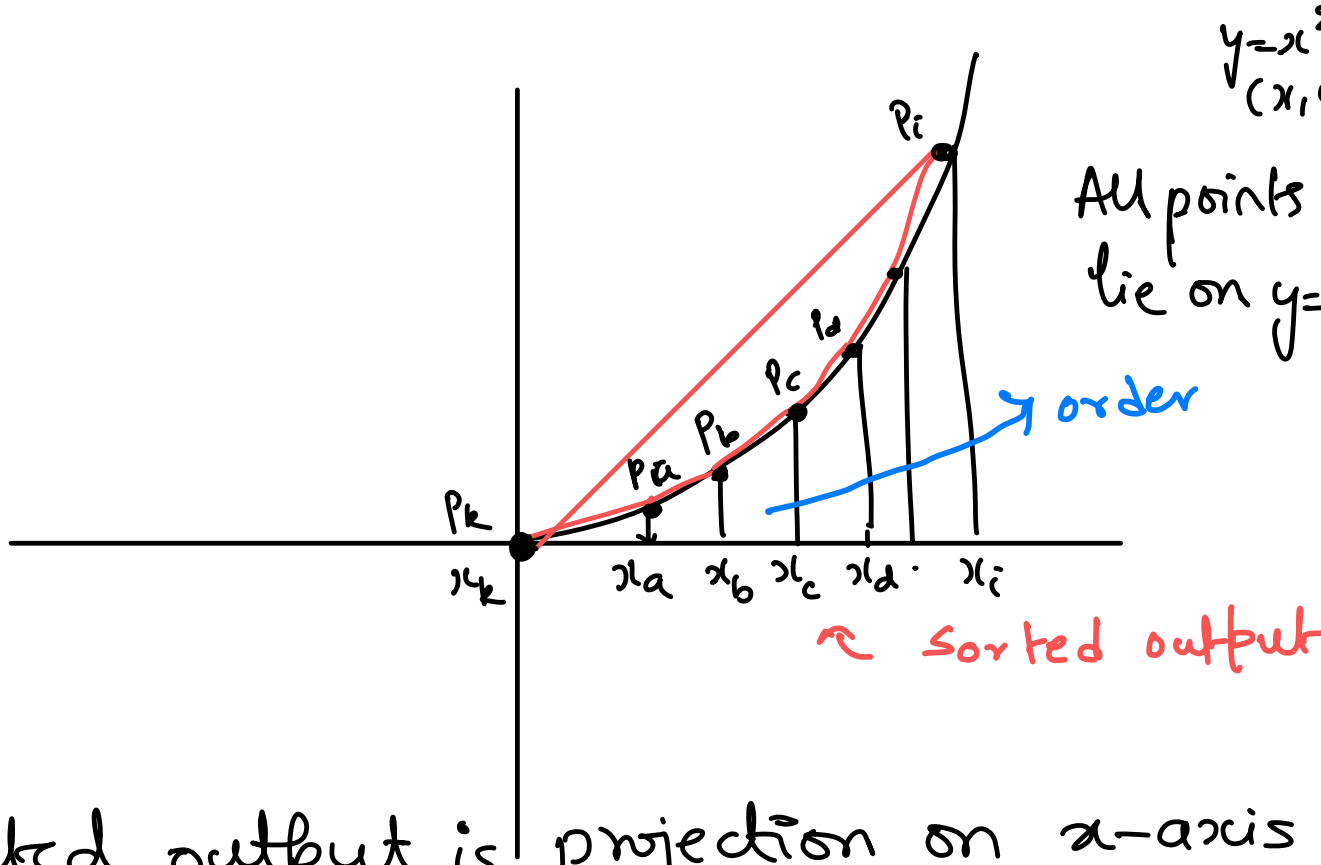
Sorting Input $A = \{x_0, x_1, \cdots, x_n\}$ (1D)

Construction: Choose (1D → 2D)

$p_0 = (x_0, x_0^2)$, $p_1 = (x_1, x_1^2)$, $p_n = (x_n, x_n^2)$

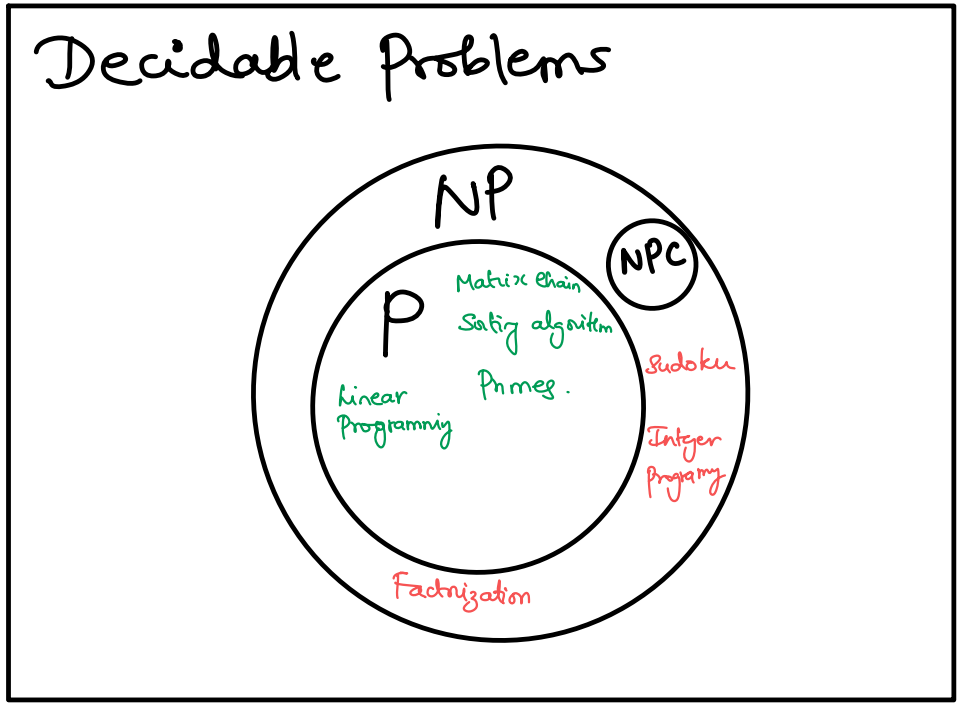
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Form convex hull of $p_0, p_1, \ldots, p_n$



Sorted output is projection on x-axis following lower part of convex hull order.

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Undecidable problems : Ex

Turing's Halting problem

P – polynomial time

NP – Non-deterministic polynomial time

(if solution is known then it can be checked in polynomial time, but method to find solution is not known)

NPC – NP Complete

Def^n : (NP Complete)

Problem A is NP Complete if

i) It is in NP ($A \in NP$)

ii) Every problem in NP reduces to it (NP-hard)

Equivalently,

$B \in NP$ and for all $A \in NP$, $A \leq_p B$

If even one NPC problem can be solved in polynomial time,

all NP problems can be solved using reductions.

This will prove that

$P = NP$.

(Millenium Problem with $1 Million prize: Is $P = NP$?)

↑ still an open problem.

P	NPC
① Shortest Path	① Longest Path
② Euler Circuit → (Cover all edges exactly once)	② Travelling Salesman Problem (Hamiltonian Circuit) (Cover all vertices exactly once)
③ Testing solution exists using invariant in n^2 puzzle. (Reachability is in P)	③ n^2-puzzle (Given K, whether atmost K step solution exists? $d(s,g) \leq K$?)

start state

goal state

Note: $d(s,g) \leq G(n)$ ↑ God's number

④ Ordinary Knight's tour on nxn chessboard (No forbidden squares) $\in P$

⑤ Ordinary N-Queens problem (No forbidden squares on nxn chessboard) $\in P$

⑥ Fixed size 9x9 Sudoku (ordinary) $\in P$

④ Generalized Knight's tour on nxn chessboard (with some forbidden squares) $\in NPC$

⑤ Generalized N-Queens with some forbidden squares. $\in NPC$

⑥ Generalized $n^2 \times n^2$ Sudoku (Can the partially filled board be completed?) $\in NPC$.

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Turing's Halting Problem:

There is no algorithm $H(P, x)$ that for every program P and input x, correctly decides whether P eventually halts on x.

Proof:

In a contrary way, suppose such an algorithm exists.

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That is,

$$H(P,x) = \begin{cases} 1 \ (\text{Yes}), & \text{if } P \text{ halts on input } x \\ 0 \ (\text{No}), & \text{if } P \text{ does not halt on input } x. \end{cases}$$

predicts surely

(I)

Turing's Construction (Look at opposite of H)

Define

$$\overline{H}(P) = \begin{cases} \text{run forever} & \text{if } H(P,P)=1 \\ \text{halts} & \text{if } H(P,P)=0 \end{cases} \quad \text{---(II)}$$

(Feed $\overline{H}$ to itself) - Reflexivity

That is,

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Choose $P = \overline{H}$ in (II).

$$\overline{H}(\overline{H}) = \begin{cases} \text{run forever} & \text{if } H(\overline{H}, \overline{H}) = 1 \quad \text{Case 2} \\ \text{halts} & \text{if } H(\overline{H}, \overline{H}) = 0. \quad \text{Case 1} \end{cases}$$

— (III)

Analyze (III):

Case 1 : $H(\overline{H}, \overline{H}) = 0$

By (I), H predicts "surely" that $\overline{H}$ does not halt on input $\overline{H}$.

That is $\overline{H}(\overline{H})$ does not halt

This contradicts (III)

which says $\overline{H}(\overline{H})$ Halts.

Case 2: $H(\overline{H}, \overline{H}) = 1$.

By (I), H surely predicts that

$\overline{H}$ halts on input $\overline{H}$

That is, $\overline{H}(\overline{H})$ halts

This contradicts (III)

which says $\overline{H}(\overline{H})$ runs forever.

Hence Halting Problem is undecidable.

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Collatz Conjecture or 3n+1 problem.

$$C(n) = \begin{cases} n/2 & n \text{ is even} \\ 3n+1 & n \text{ is odd} \end{cases}$$

$n = 6$.

$6 \to 3 \to 10 \to 5 \to 16 \to 8 \to 4 \to 2 \to 1$.

$n = 3$: $3 \to 10 \to 5 \to 16 \to 8 \to 4 \to 2 \to 1$

$n = 4$: $4 \to 2 \to 1$

$n = 5$: $5 \to 16 \to 8 \to 4 \to 2 \to 1$

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Starting n	Steps to reach 1	Largest value reached
2	1 (2 terms)	2
3	7 (8 terms)	16
4	2 (3 terms)	4
5	5 (6 terms)	16
6	8 (9 terms)	16
4.6×10^{19}	2480 (2481 terms)	1.18×10^{25}

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(Modf of Turings Halting Prob)

Post Correspondence Problem

Let $D_1 = \begin{bmatrix} u_1 \\ v_1 \end{bmatrix}$, $D_2 = \begin{bmatrix} u_2 \\ v_2 \end{bmatrix}$, $\ldots D_n = \begin{bmatrix} u_n \\ v_n \end{bmatrix}$.

u_i, v_i are in alphabet Σ.

The problem asks whether there exists a sequence $(i_1, i_2, \ldots, i_k)$ such that

(Concatenation of top) $u_{i_1} u_{i_2} \cdots u_{i_k}$ = (Concatenation of bottom) $v_{i_1} v_{i_2} \cdots v_{i_k}$

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- A tile can be selected multiple times.

eg: $D_1 = \begin{bmatrix} a \\ ab \end{bmatrix}$, $D_2 = \begin{bmatrix} ba \\ a \end{bmatrix}$

$\overset{i_1, i_2}{(1, 2)}$ ← Solution of PCP

Top: $a(ba)$

Botton: $(ab)a$

$abba = a(ba) = (ab)a$

② $P_1 = \begin{bmatrix} a \\ ab \end{bmatrix}$ $P_2 = \begin{bmatrix} b \\ c \end{bmatrix}$, $P_3 = \begin{bmatrix} ca \\ a \end{bmatrix}$

$(1, 2, 3) \longleftarrow$ Soln of PCP for ②

Top: $(a)(b)(ca)$

Bottom: $(ab)(c)(a)$

$abca = (a)(b)(ca) = (ab)(c)(a)$

③ $P_1 = \begin{bmatrix} aa \\ a \end{bmatrix}$ $P_2 = \begin{bmatrix} a \\ aaab \end{bmatrix}$ $P_3 = \begin{bmatrix} bbb \\ bbaa \end{bmatrix}$

$\Sigma = \{a, b, c\}$

(Shortcut) Solution: $(1, 1, 2, 3, 1, 1)$

Top: $(aa)(aa)(a)(bbb)(aa)(aa)$

Bottom: $(a)(a)(aaab)(bbaa)(a)(a)$

$=$

$=$ $aaaaabbbaaaa$

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The above solution is shortest

Used: BFS with pruning

eg: $D_1 = \begin{bmatrix} b \\ bba \end{bmatrix}$ $D_2 = \begin{bmatrix} aa \\ a \end{bmatrix}$ $D_3 = \begin{bmatrix} bab \\ baa \end{bmatrix}$

BFS Level 1: (length 1)

Keep a seq^n only if one row is a prefix of other

Sequence	Top	Bottom	Decision (Keep or not)
(1)	b	bba	keep (✓) (b is prefix of bba)
(2)	aa	a	keep (✓)
(3)	bab	baa	Prune (X)

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Retain (1), (2).

BFS Level 2: (Length 2)

Sequence	Top	Bottom	Decision
(1, 1)	bb	bbabba	keep ✓
(1, 2)	baa	bbaa	✗
(1, 3)	bbab	bbabaa	✓
(2, 1)	aab	abba	✗
(2, 2)	aaaa	aa	✓
(2, 3)	aabab	abaa	✗

BFS Level 3 (Length 3)

Sequence	Top	Bottom	Decision
(1,1,1)	bbb	bbabbabba	×
(1,1,2)	bbaa	bbabbaaa	×
(1,1,3)	bbbab	bbabbabaa	×
(1,3,1)	bbabb	bbabaabba	×
(1,3,2)	bbabaa	bbabaaa	✓
(1,3,3)	bbabbab	bbabaabaa	×
(2,2,1)	aaaab	aabba	×
(2,2,2)	aaaaa	aaa	✓

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(2,2,3)	aaaabab	aabaa	X

Retain (1,3,2) and (2,2,2).

BFS Level 4 : (Length 4)

Sequence	Top	Bottom	Decision
(1,3,2,1)	bbabaab	bbabaaabba	X
(1,3,2,2)	bbabaaaa	bbabaaaaa	Match
(1,3,2,3)	bbabaabab	bbabaaabaa	X
(2,2,2,1)	aaaaab	aaabba	X

Shortest PCP

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$(2,2,2,2)$	$aaaaaaaa$	$aaaa$	$\times$
$(2,2,2,3)$	$aaaaabab$	$aaabaa$	$\times$

Note: PCP is undecidable.

Proof Hint:

Halting $\leq$ PCP (Reduction many to 1)

↑

Undecidable (proved)

$\Rightarrow$ PCP is at least as hard as halting problem

$\Rightarrow$ PCP is undecidable.